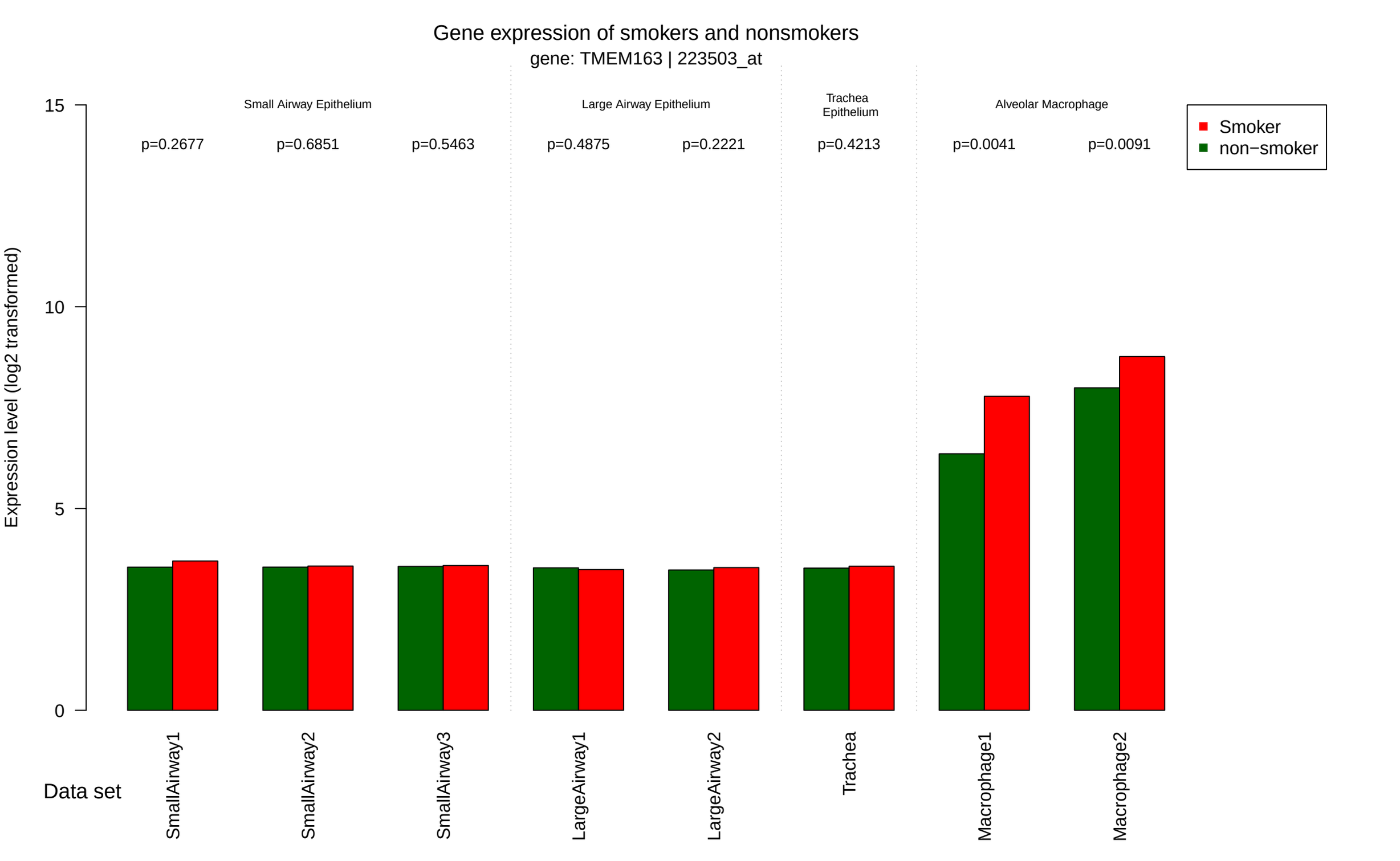
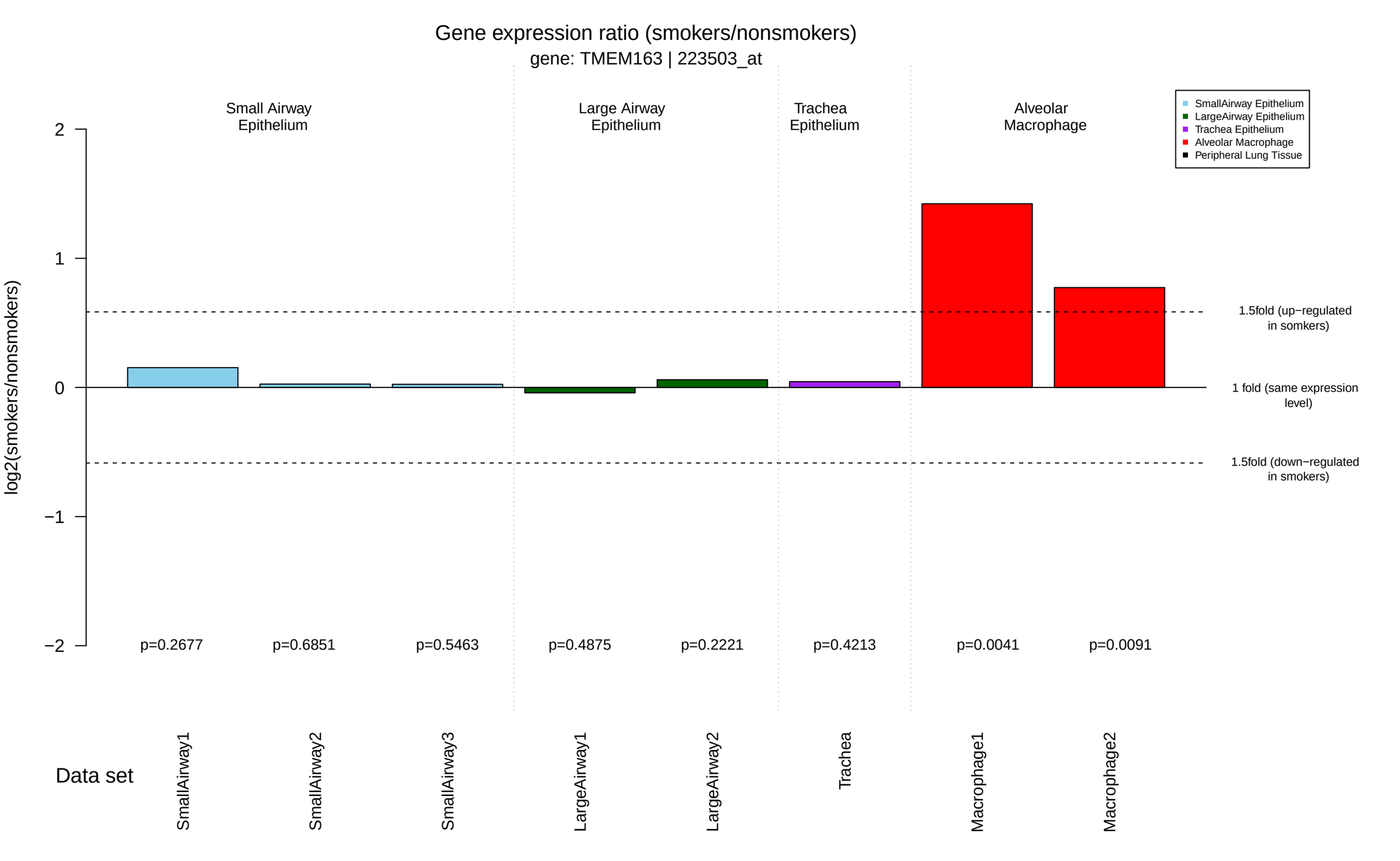
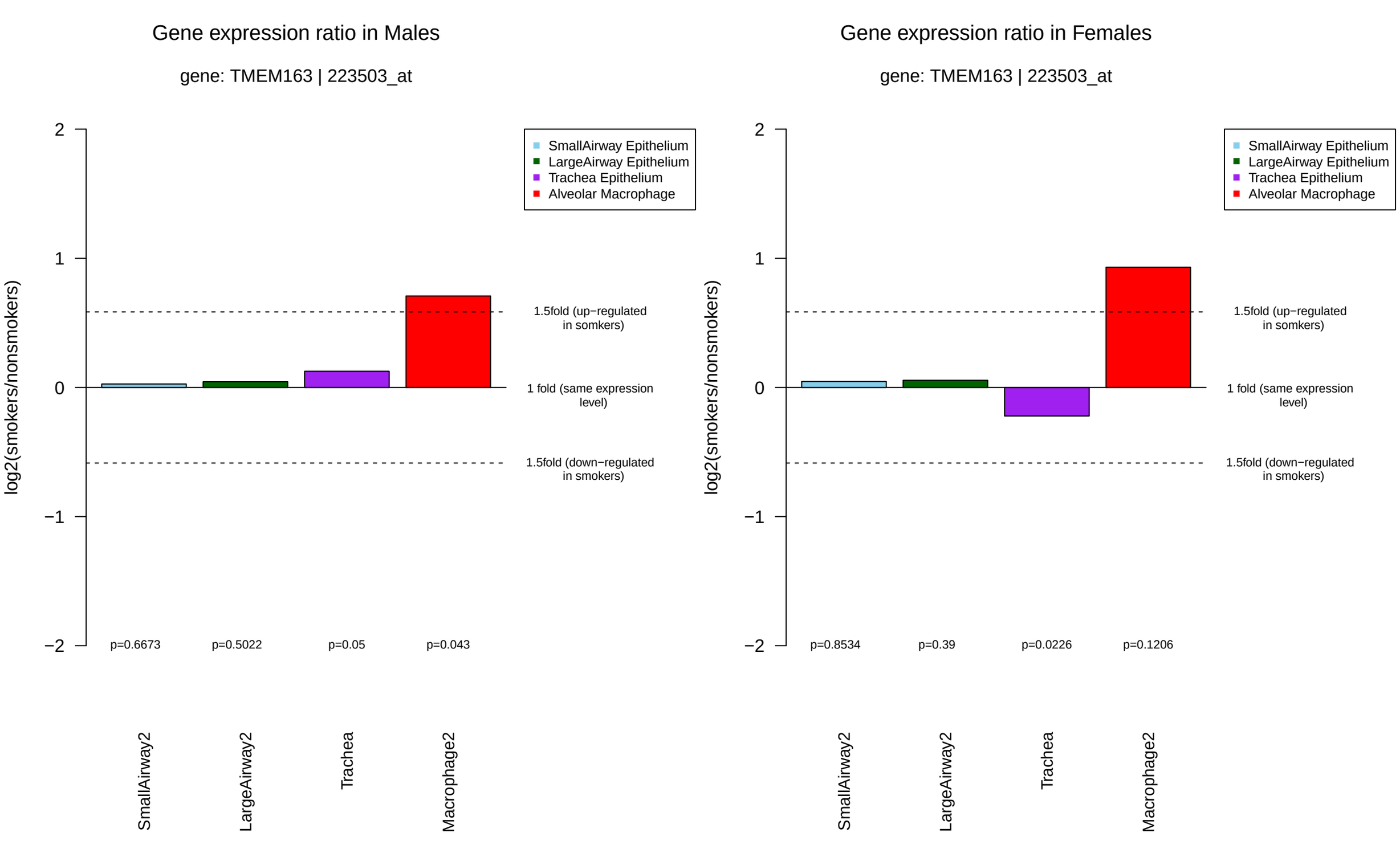
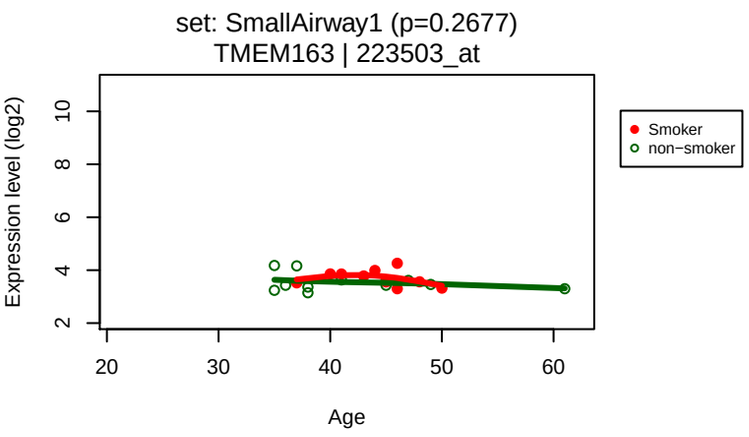
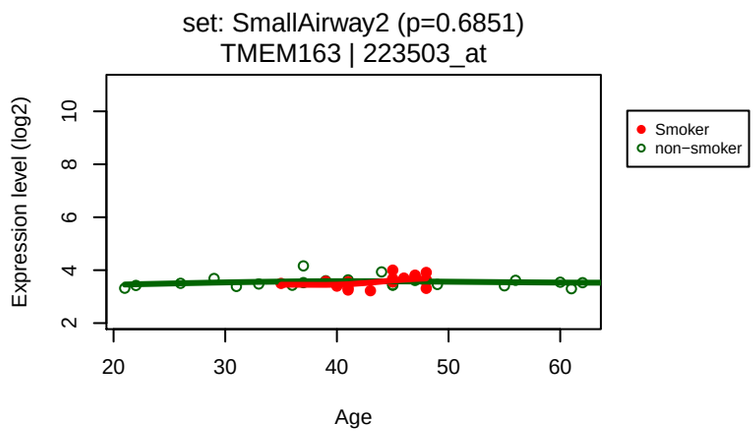
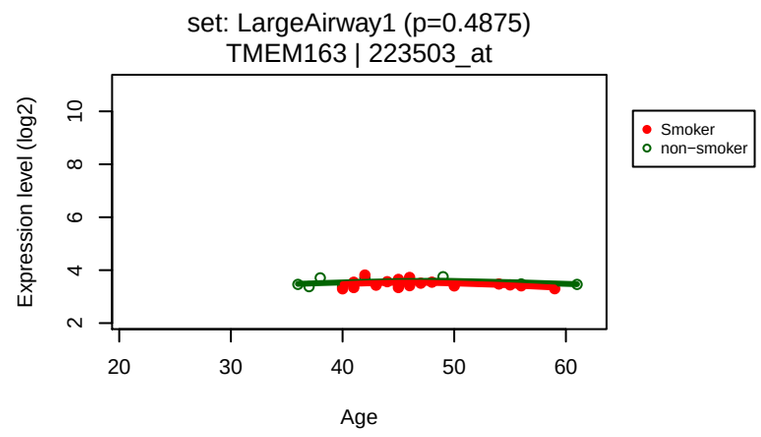

Gene expression of smokers and nonsmokers
gene: TMEM163 | 223503_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2677
p=0.6851
p=0.5463
p=0.4875
p=0.2221
p=0.4213
p=0.0041
p=0.0091
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

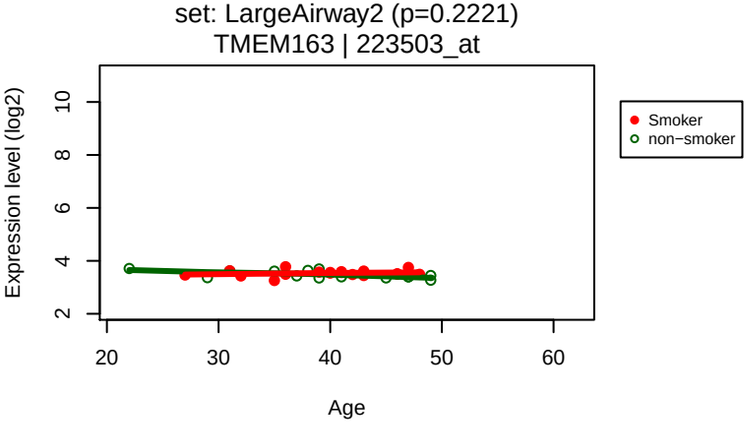

Gene expression ratio (smokers/nonsmokers)
gene: TMEM163 | 223503_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2677
p=0.6851
p=0.5463
p=0.4875
p=0.2221
p=0.4213
p=0.0041
p=0.0091
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

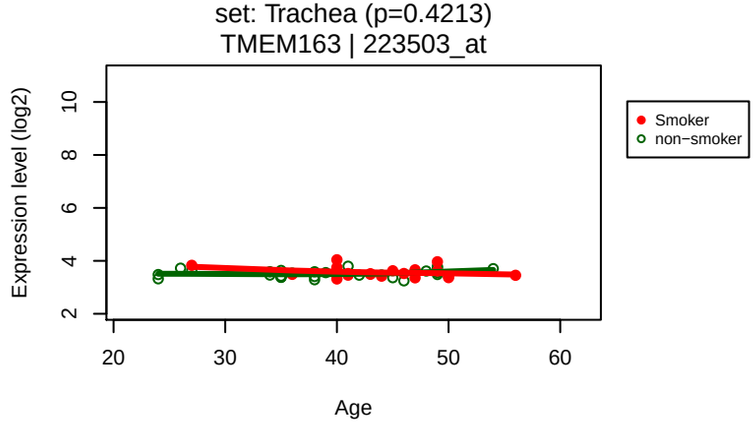

Gene expression ratio in Males
gene: TMEM163 | 223503_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6673
p=0.5022
p=0.05
p=0.043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM163 | 223503_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8534
p=0.39
p=0.0226
p=0.1206
SmallAirway2
LargeAirway2
Trachea
Macrophage2

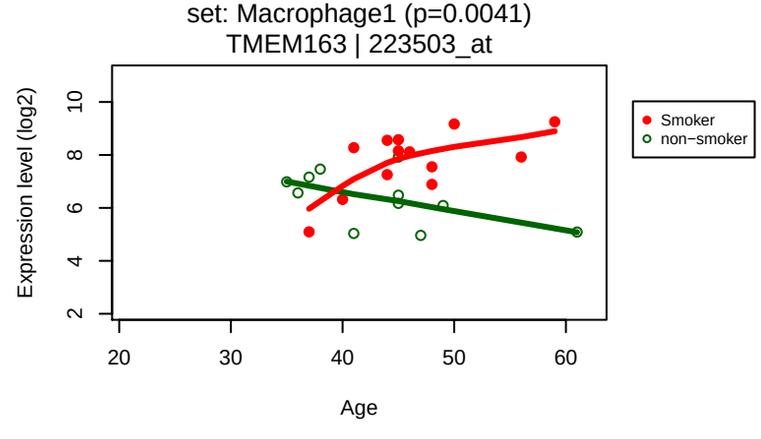

set: SmallAirway1 (p=0.2677)
TMEM163 | 223503_at
Expression level (log2)
Age
Smoker
non−smoker

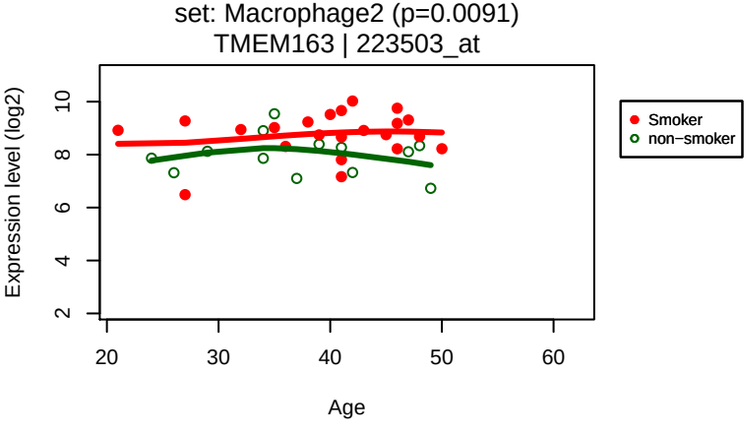

set: SmallAirway2 (p=0.6851)
TMEM163 | 223503_at
Expression level (log2)
Age
Smoker
non−smoker

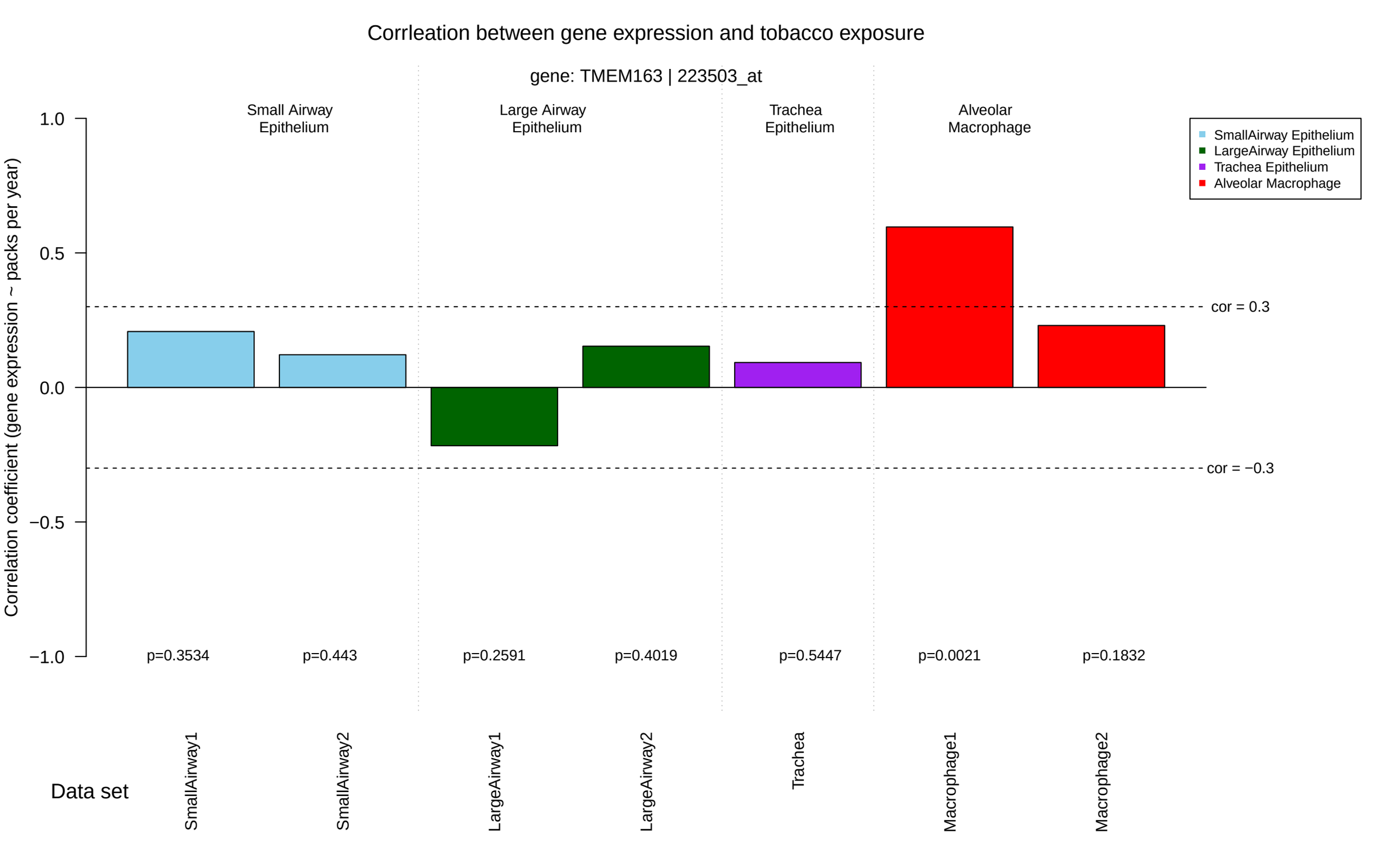

set: LargeAirway1 (p=0.4875)
TMEM163 | 223503_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2221)
TMEM163 | 223503_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4213)
TMEM163 | 223503_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0041)
TMEM163 | 223503_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0091)
TMEM163 | 223503_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM163 | 223503_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3534
p=0.443
p=0.2591
p=0.4019
p=0.5447
p=0.0021
p=0.1832
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2